PMS Assignment - 1

1) governing equation $\rho C_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k(T)\frac{\partial T}{\partial x} \right)$

$k(T)$: thermal conductivity of silver

Boundary conditions

$x=0 \qquad T = T_0\,(2+\sin(2\pi\omega t))$

$x=L \qquad T = T_0$

Initial condition

$t=0 \qquad T = T_0 = 300K$

$L = 0.01m$

$\rho = 10{,}800\ kg/m^3$

$C_p = 238.65\ J/kg\cdot K$ } given in Perry's Handbook

Let us use the following 5 values of $\omega \rightarrow$ 1HZ, 5HZ, 10HZ, 20HZ, 50HZ

Number of time points $N=10$

For $\omega = 1HZ$,

To include 5 full cycles, time of simulation = 5 seconds

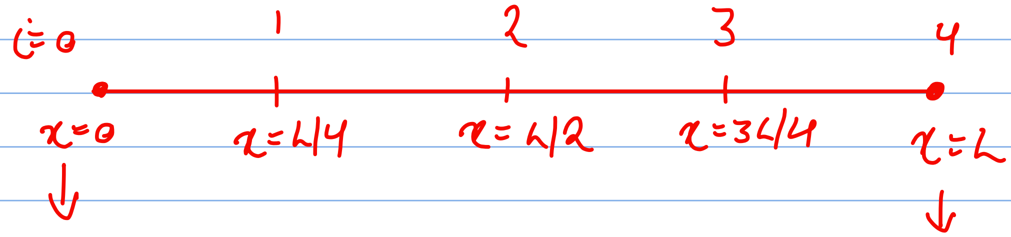
$\Delta t = \frac{5}{1\times 10} = 0.5s$

$\Delta x = \frac{10^{-2}}{4} = 0.0025m \Rightarrow 2.5\times 10^{-3} m$

From online data points, we get the following

$K(T) = 10^{-5}T^3 - 8.44\times 10^{-3}T^2 + 2.591T + 136.63855$

$K(T) = 3\times 10^{-5}T^2 - 16.88\times 10^{-3}T + 2.591$



$T = T_0(2+\sin(2\pi\omega t)$ $\qquad$ $T = T_0$

a) Explicit Method

$$\frac{\partial T}{\partial t} = \frac{T_i^{n+1} - T_i^{n}}{\Delta t} \quad \{\text{Forward difference}\}$$

$$\frac{\partial T}{\partial x} = \frac{T_{i+1}^{n} - T_i^{n}}{\Delta x}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{T_{i+1}^{n} - 2T_i^{n} + T_{i-1}^{n}}{(\Delta x)^2}$$

Now, $\rho c_p \dfrac{\partial T}{\partial t} = \dfrac{\partial}{\partial x}\left(k(T)\dfrac{\partial T}{\partial x}\right)$

$$\frac{\partial T}{\partial t} = \alpha\left[k'(T)\left(\frac{\partial T}{\partial x}\right)^2 + k(T)\frac{\partial^2 T}{\partial x^2}\right] \quad \text{where } \alpha = \frac{1}{\rho c_p}$$

$$\frac{T_i^{n+1} - T_i^{n}}{\Delta t} = \alpha\left[K'(T_i^{n})\left[\frac{T_{i+1}^{n} - T_i^{n}}{\Delta x}\right]^2 + k(T_i^{n})\frac{T_{i+1}^{n} - 2T_i^{n} + T_{i-1}^{n}}{(\Delta x)^2}\right]$$

$$\therefore \quad T_i^{n+1} = \frac{\alpha\Delta t}{\Delta x^2}\left[K'(T_i^{n})(T_{i+1}^{n} - T_i^{n})^2 + K(T_i^{n})(T_{i+1}^{n} - 2T_i^{n} + T_{i-1}^{n})\right] + T_i^{n}$$

From our code, we found $N=10$ (This soln does not converge)

$\therefore$ we take some higher value of N for convergence

$\therefore$ ω can be changed for different results

b) Implicit Method

Equation: $T_i^{n+1} = T_i^n + \lambda \left[K'(T_i^n)(T_{i+1}^{n+1} - T_i^{n+1}) + K(T_i^n)[T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}] \right]$

$K(T) : p \qquad K'(T) : p'$

$$T_i^{n+1} = T_i^n + \lambda \left[p'(T_{i+1}^{N+1} - T_i^{N+1}) + a\,(T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}) \right]$$

$T_i^n : c_i$ & $T = [T_1 \ T_2 \ T_3]^T$

$$\begin{bmatrix} -(1+\lambda p' + 2\lambda p & \lambda(p'+p) & 0 \\ \lambda p & -(1+\lambda p' + 2\lambda p) & \lambda(p'+p) \\ 0 & \lambda p & -(1+\lambda p' + 2\lambda p) \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

Temperature profile To be simulated

c) Crank Nicholson Method

Average of temperature profile obtained through implicit and explicit method

d) Since, thermal conductivity 'k' is not constant and varying with temperature, it is not possible to apply Von-Neumann stability criterion

Temperature values explode to very high values (solution does not converge) as seen from the graph in the code for $\omega = 1$ HZ and small values of N [1, 10, 100 HZ etc.]

Von-Neumann stability criterion $\frac{\alpha \Delta t}{\Delta x^2} \leq 0.5$

N : # time points

a) Any value of w } Solution of implicit scheme converges
Any value of N } to steady state solution

The same is not true for explicit scheme

b) For large values of N } both schemes converge

c) For small values of N } Implicit is more accurate than explicit scheme (assuming convergence in both cases)

e) To solve N-system of equations in implicit scheme, different methods, which can be followed

→ Gauss elimination method $O(N^3)$
→ Gauss Newton Method $O(N^2)$
→ Thomas Algorithm $O(N)$

f) When values of w are changed, a phase difference b/w average temperature response is observed at $x=0$

$w \rightarrow$ 1 HZ, 10 HZ, 20 HZ, 50 HZ

$W > 15$ HZ → less fluctuation in temperature profile at $x=L$

$T = \frac{1}{w}$ (As $w\uparrow$ $T\downarrow$)

Effect of initial temperature (T_0) at $x=L$ will be pronounced as the time of observation decreases

Critical value will depend on $\alpha(\rho, c_p)$ of w